Monklap

A touchless ambient scent system that answers a clap.

TYPE
Wall-mounted / desktop IoT object

TOOLS
Rhino 8
Bambu Studio
ESP32

DESIGNER
Andy Oh

Scent, on cue

Monklap is a **wall-mounted or desktop** diffuser shaped like a friendly monkey. Walk up and clap, and it responds with a pulse of fragrant mist from its mouth.

No buttons, no app, nothing to touch.

01

Touchless

Sound and proximity replace the spray trigger, for moments when your hands are busy, wet or full.

02

Interactive

An OLED face gives it a character and shows visible feedback for every state.

03

Ambient

A wick-fed ultrasonic atomizer releases a 1.5-second burst of mist on demand, not a constant stream.

FRONT VIEW — FINAL PROTOTYPE

Monklap is for...

Three groups who want scent to feel like a moment of play, not another chore.

01

Aesthetic & tech enthusiasts

Gen Z and millennials looking for desk-friendly home objects with sound-driven, physical interaction rather than another standard IoT device.

DESK SETUPS | CHARACTER OBJECTS

02

Touchless, functional users

People who need instant, hands-free odor control while cooking, caring for pets, or whenever their hands are busy.

KITCHENS | PETS | BUSY HANDS

03

Experiential commercial spaces

Pop-up stores, cafés and galleries that want interactive, sensory branding at the entrance — a scent that greets visitors.

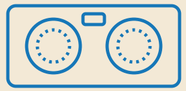
POP-UPS | CAFÉS | GALLERIES

Two sensors, two outcomes

INPUT · SENSE

Ultrasonic sensor

NOSTRILS



Checks that someone is near

SOMEONE NEAR?

INPUT · SENSE

Sound sensor

LOWER LEFT

Listens for a clap or noise

CLAP / NOISE

CONTROLLER

ESP32

Reads both sensors, picks a face, decides when to mist

USB 5V

FACE FRAMES

OUTPUT · RESPOND

2.8″ OLED display

EYES

Draws the face state

ON / OFF

DRIVER BOARD

OUTPUT · RESPOND

Ultrasonic atomizer

MOUTH

Fires a 1.5 s burst of mist

INTERACTION LOGIC

01 Rest

Nobody near, or no sound

→ SCREEN DIMMED

02 Wake

Someone near + clap or noise

→ EYES OPEN

03 Smile

Recognizes the clap

→ SMILES BACK

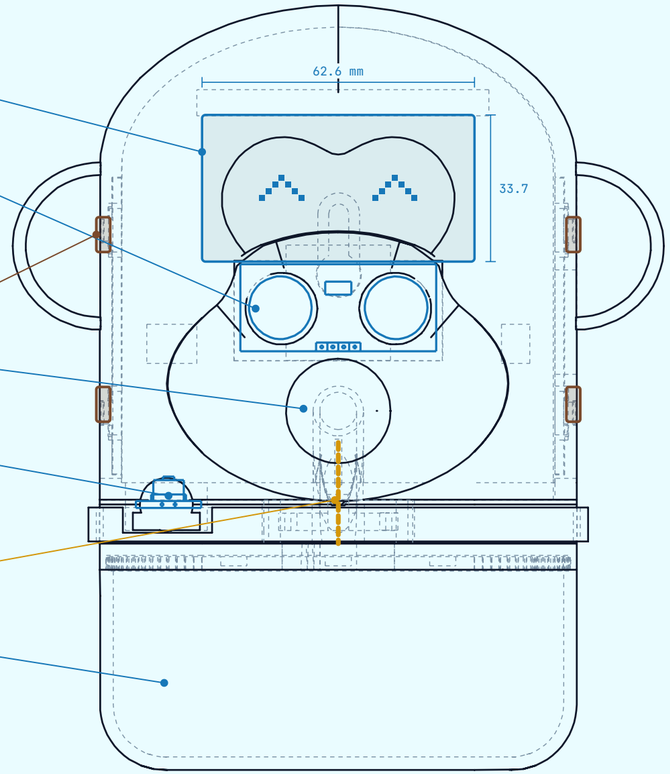
04 Pulse

Atomizer on 1.5 s

→ THEN BACK TO REST

Wireframe breakdown

FRONT



2.8″ OLED display
EYES · 62.6 × 33.7 MM

Ultrasonic sensor
NOSTRILS

Snap-fit hooks ×4
SIDE WALLS

Ultrasonic atomizer
MOUTH

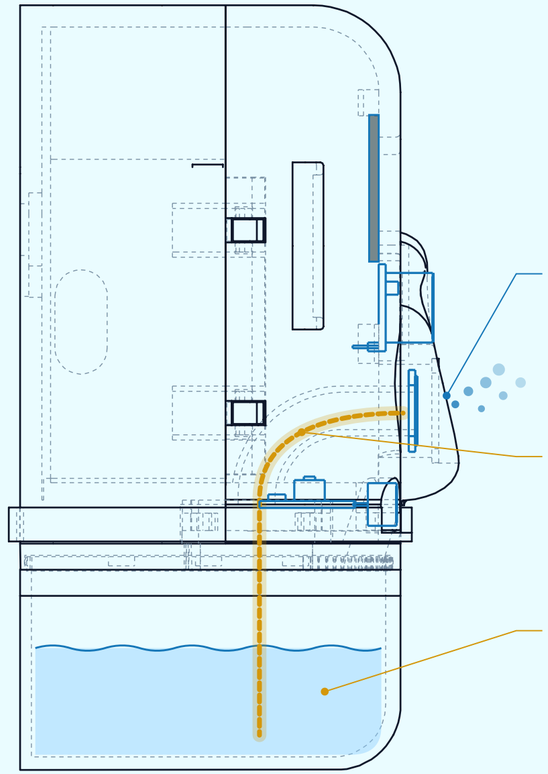
Sound sensor
LOWER LEFT

Wick channel
BAYONET → MOUTH

Bayonet tank
LID + TUB, ONE PIECE

HIDDEN LINES ON — DRAWN FROM THE RHINO MODEL

SIDE — WATER PATH



Mist out
ATOMIZER AT THE MOUTH

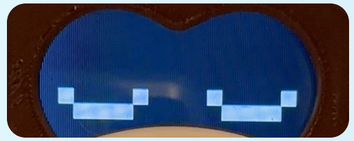
J-shaped wick
WIDEST BEND THAT FITS

Wick draws water up
BY CAPILLARY ACTION

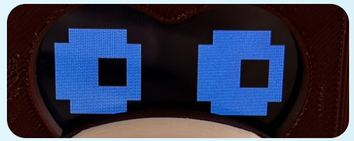
FACE →

● ELECTRONICS ● WATER PATH ● STRUCTURE

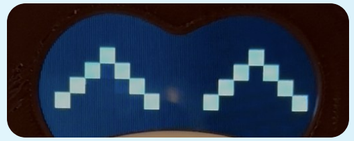
FACE STATES — REAL SCREEN



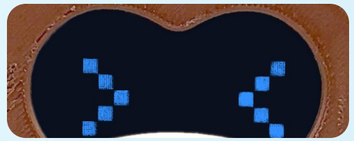
01 **Rest**
Nobody near, or no sound



02 **Wake**
Someone near + clap or noise

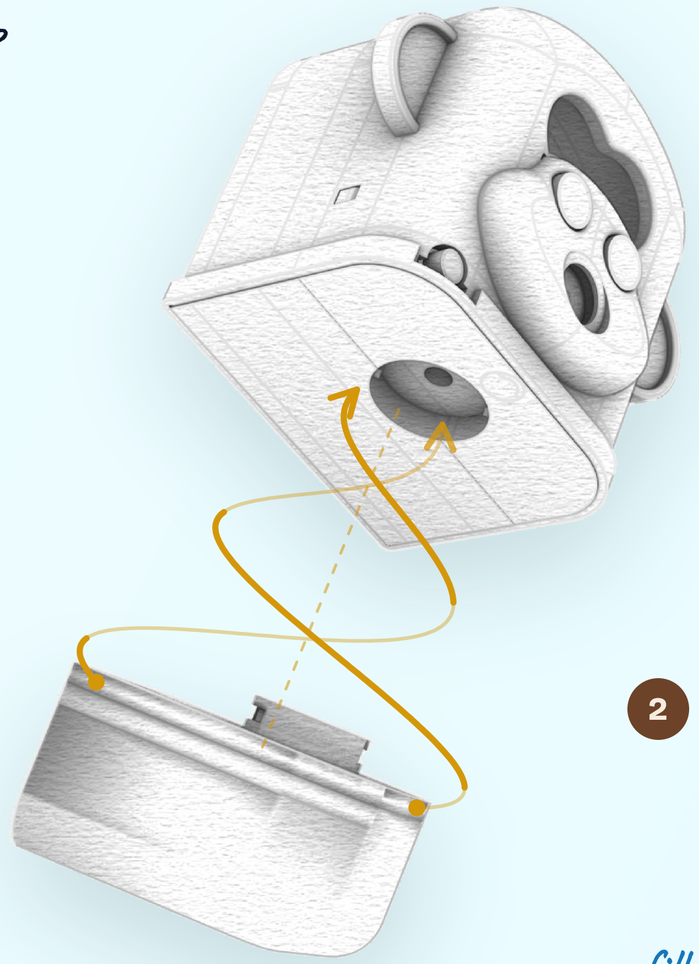


03 **Smile**
Smiles back at the clap



04 **Pulse**
1.5 s burst of mist

3 hang & clap



2 twist to lock

1 fill the tank with water

Fill, twist, clap.

01 **Fill the tank**

Fill the tank with water and a water-soluble air freshener.

02 **Twist into the head**

The lid's bayonet locks into the head. A wick carries water up to the mouth.

03 **Hang it and clap**

A keyhole slot on the back hangs it on a single screw. No buttons. The face does the rest.

WHAT'S INSIDE THE FACE

Eyes · OLED display

Mouth · ultrasonic atomizer

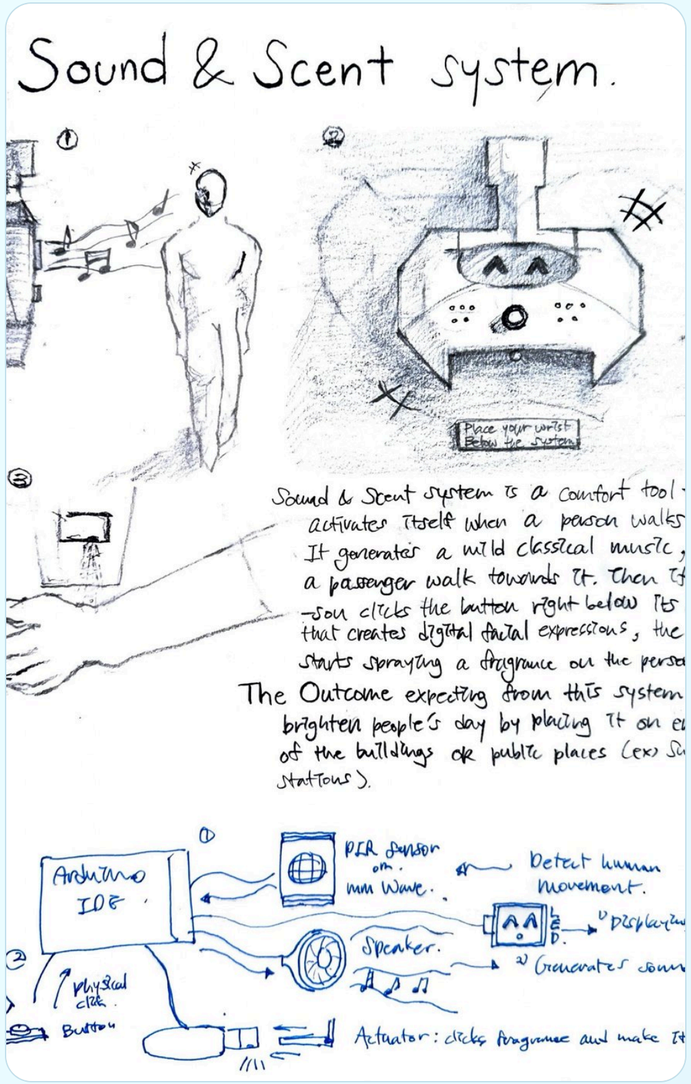
Nostrils · ultrasonic sensor

Lower left · sound sensor

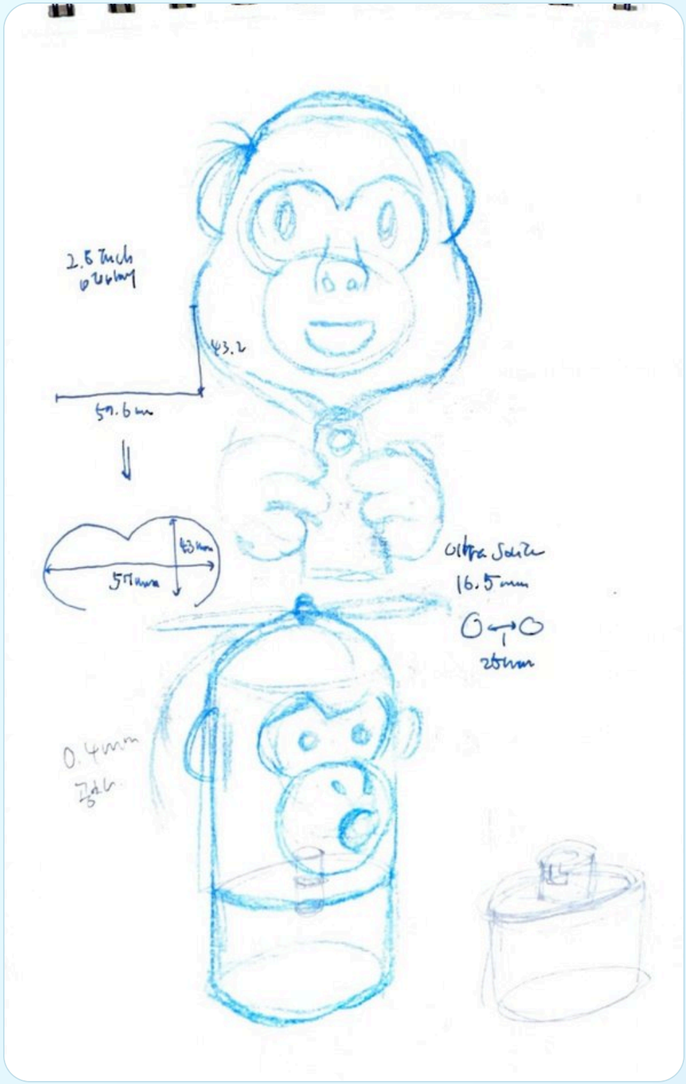
From a button to a clap

Monklap began as a “Sound & Scent System”: a comfort tool for public spaces that played music as people walked by, then sprayed fragrance on their wrists at the press of a button.

KEPT	Walk-up wake, a digital face, scent as a reward
CHANGED	The button became a clap; the form became a monkey

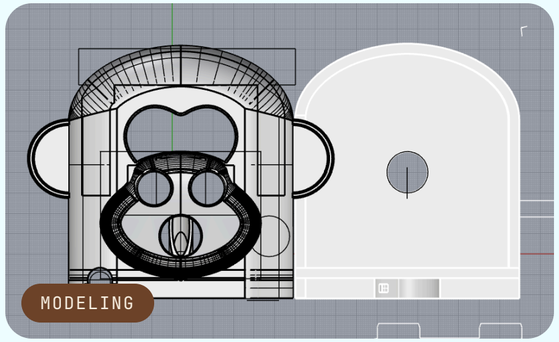


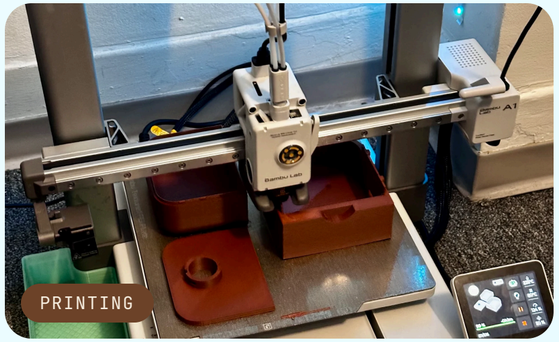
INITIAL CONCEPT — WALK-BY MUSIC & WRIST SPRAY

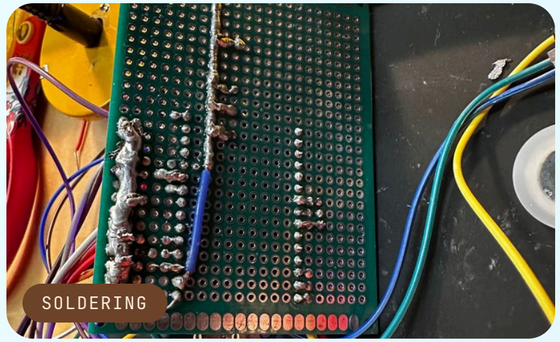


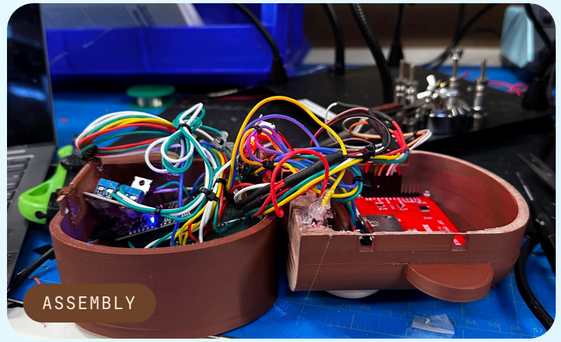
FORM STUDIES — HEAD, SENSORS & TANK

Process & lessons









Modeling

Taking Monklap from rough sketches to 3D models in Rhino taught me that the more specific a plan is, the less time I lose troubleshooting changes later. I now understand not only how to model objects for real-world use, but also how to plan a project so each stage of making flows into the next with less friction.

Soldering

I troubleshot shorts caused by broken cords, cold solder joints and tangled stripped wires. Each time, I reopened the board, resoldered the joints and added hot glue so the same problem wouldn't happen again.

3D printing

I printed Monklap's parts from my Rhino models on a Bambu Lab A1 using Bambu Studio. Printing itself went smoothly, but some edges warped, so I had to sand those surfaces flat.

Assembly

Using the joint structures I designed during modeling, I fitted the sensors, display and atomizer in order. My plans for the circuits and wick didn't always match reality, so it took a lot of trial and error. After four rounds of printing and fitting, I arrived at the current Monklap.

09 — FINAL PROTOTYPE

Monklap, awake.

The finished Monklap: live OLED eyes, ultrasonic sensors in the nostrils and an atomizer in the mouth, all running on an ESP32.

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FRONT — AWAKE

THREE-QUARTER VIEW

CLAP → PULSE OF MIST